

Using the set operations

1) Create a matrix query to display the job, the salary for that job based on department number, and the total salary for that job, for departments 10, 20, and 30, giving each column an appropriate heading.

```
SELECT JOB,  
       DEPTNO,  
       SUM(SAL)  
FROM EMP  
WHERE DEPTNO IN (10,20,30)  
GROUP BY JOB, DEPTNO  
ORDER BY JOB, DEPTNO;
```

Output:-

```
SQL> SELECT JOB,  
2      DEPTNO,  
3      SUM(SAL)  
4  FROM EMP  
5  WHERE DEPTNO IN (10,20,30)  
6  GROUP BY JOB, DEPTNO  
7  ORDER BY JOB, DEPTNO;
```

JOB	DEPTNO	SUM(SAL)
ANALYST	20	5600
CLERK	10	3400
CLERK	20	1900
CLERK	30	950
MANAGER	10	2450
MANAGER	20	2975
MANAGER	30	2850
PRESIDENT	10	5000
SALESMAN	30	5600

9 rows selected.
SQL>

2) Using set operator display the DEPTNO, SUM(SAL) for each dept, JOB, SUM(SAL) for each Job and Total Salary.

```
SELECT TO_CHAR(DEPTNO), TO_CHAR(SUM(SAL))  
FROM EMP  
GROUP BY DEPTNO  
UNION ALL  
SELECT JOB, TO_CHAR(SUM(SAL))  
FROM EMP  
GROUP BY JOB  
UNION ALL  
SELECT 'TOTAL', TO_CHAR(SUM(SAL))  
FROM EMP;
```

Output:-

```

SQL> SELECT TO_CHAR(DEPTNO), TO_CHAR(SUM(SAL))
2 FROM EMP
3 GROUP BY DEPTNO
4
SQL> UNION ALL
SP2-0042: unknown command "UNION ALL" - rest of line ignored.
SQL>
SQL> SELECT JOB, TO_CHAR(SUM(SAL))
2 FROM EMP
3 GROUP BY JOB
4
SQL> UNION ALL
SP2-0042: unknown command "UNION ALL" - rest of line ignored.
SQL>
SQL> SELECT 'TOTAL', TO_CHAR(SUM(SAL))
2 FROM EMP;

'TOTA TO_CHAR(SUM(SAL))
-----
TOTAL 30725

SQL>
SQL> SELECT TO_CHAR(DEPTNO), TO_CHAR(SUM(SAL))
2 FROM EMP
3 GROUP BY DEPTNO
4 UNION ALL
5 SELECT JOB, TO_CHAR(SUM(SAL))
6 FROM EMP
7 GROUP BY JOB
8 UNION ALL
9 SELECT 'TOTAL', TO_CHAR(SUM(SAL))
10 FROM EMP;

TO_CHAR(DEPTNO)
-----
TO_CHAR(SUM(SAL))
-----
20
18475

30
9400

10
18850

TO_CHAR(DEPTNO)
-----
TO_CHAR(SUM(SAL))
-----
ANALYST
5600

SALESMAN
5600

CLERK
6250

TO_CHAR(DEPTNO)
-----
TO_CHAR(SUM(SAL))
-----
MANAGER
8275

PRESIDENT
5800

TOTAL
30725

9 rows selected.

SQL>

```

3) Using Set Operator display the JOB and Deptno in employees working in deptno 20, 10, 30 in that order.

```

SELECT JOB, DEPTNO
FROM EMP
WHERE DEPTNO = 20
UNION ALL
SELECT JOB, DEPTNO
FROM EMP
WHERE DEPTNO = 10
UNION ALL
SELECT JOB, DEPTNO
FROM EMP
WHERE DEPTNO = 30;

```

Output:-

```
SQL> SELECT JOB, DEPTNO
2 FROM EMP
3 WHERE DEPTNO = 20
4 UNION ALL
5 SELECT JOB, DEPTNO
6 FROM EMP
7 WHERE DEPTNO = 10
8 UNION ALL
9 SELECT JOB, DEPTNO
10 FROM EMP
11 WHERE DEPTNO = 30;
```

JOB	DEPTNO
ANALYST	20
CLERK	20
MANAGER	20
CLERK	20
ANALYST	20
CLERK	10
MANAGER	10
CLERK	10
PRESIDENT	10
SALESMAN	30
MANAGER	30

JOB	DEPTNO
SALESMAN	30
SALESMAN	30
SALESMAN	30
CLERK	30

15 rows selected.

SQL>